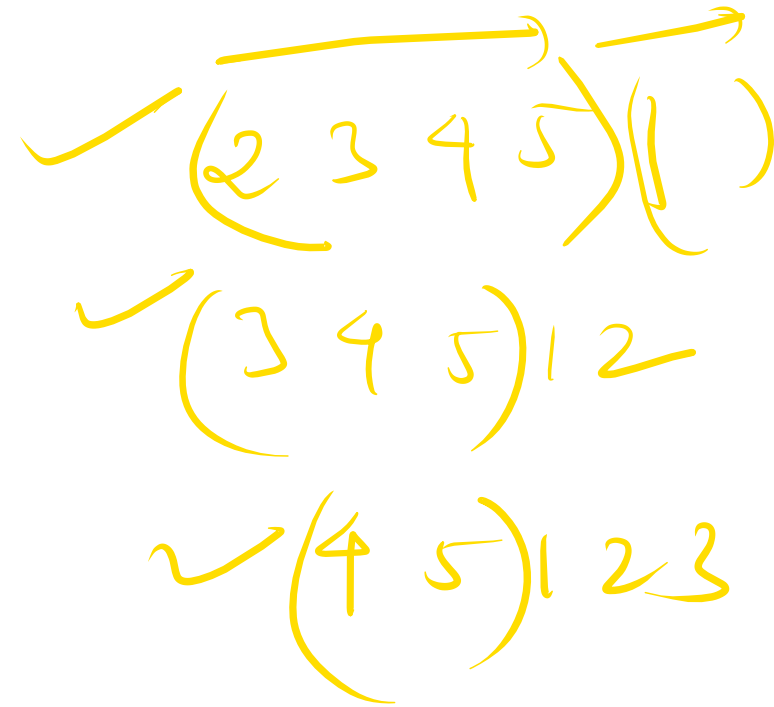
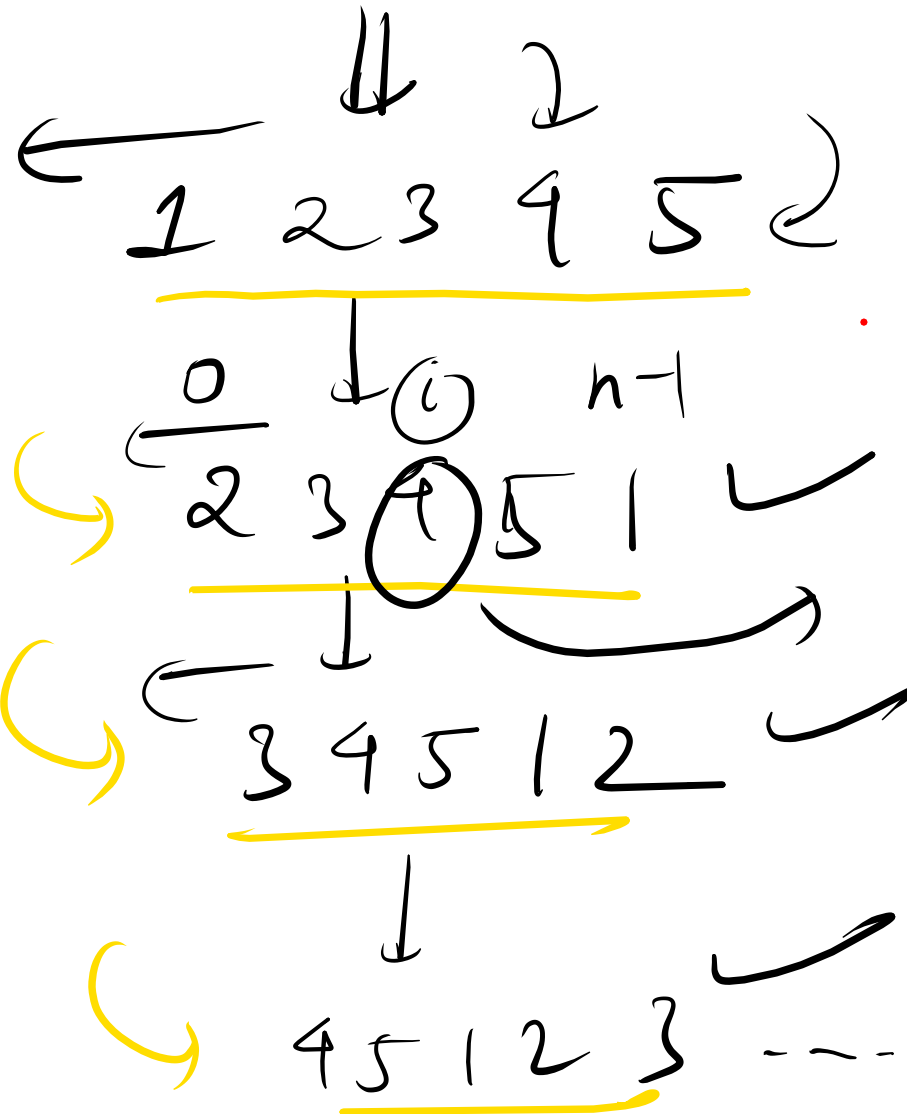


✓ sorted Rotated array?

left rotation

$n \rightarrow$



BS

$$\begin{array}{l} \overrightarrow{(2\ 3\ 4\ 5)} \overrightarrow{(1)} \checkmark \\ (3\ 4\ 5)(1\ 2) \checkmark \\ (4\ 5)(1\ 2\ 3) \checkmark \end{array} \quad ?$$

$$\begin{array}{cc} (s_1) & (s_2) \\ \hline \uparrow & \uparrow \end{array}$$

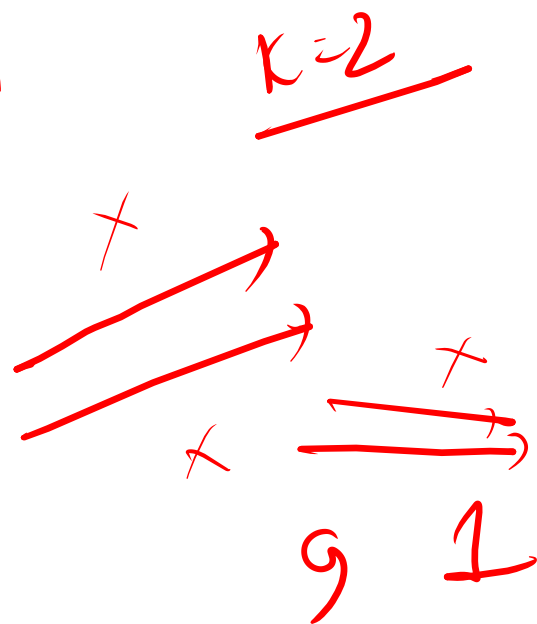
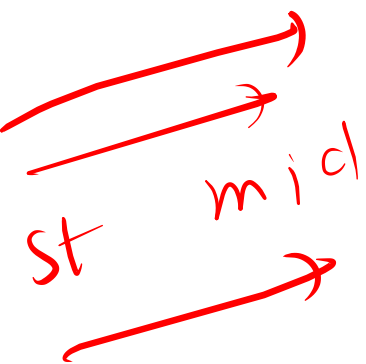
US 221
~~sorted~~ $\rightarrow$ low high
 low high

st mid
 (5 7) (8) (9 1 2)
 X X

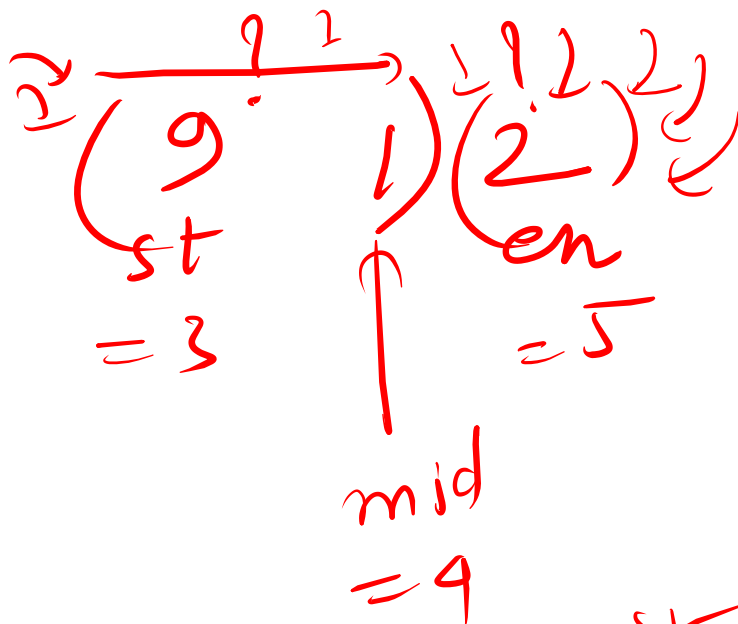
K=2

low
 K
 us//
 K

st mid
 $arr[st] \leq K \leq arr[mid]$
 $5 \leq 2 \leq 8$



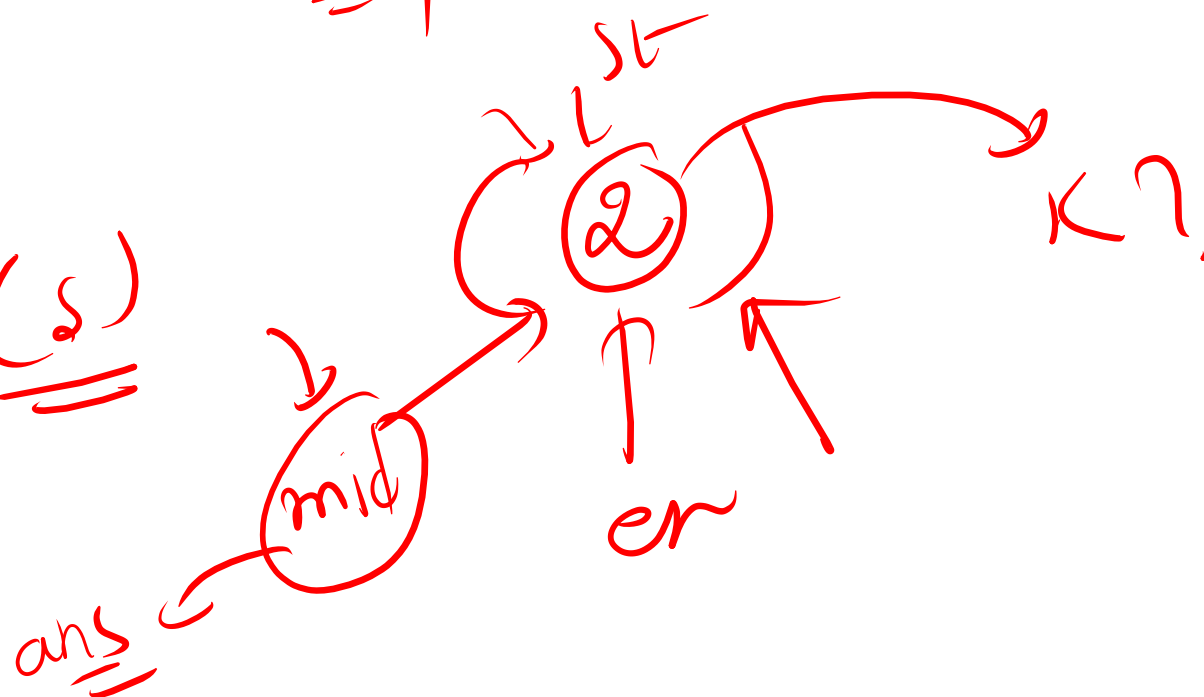
(vs) (s)



$$k=2?$$

$$\frac{mid+1}{en}$$

$$\underline{2} <= \underline{k=2} <= \underline{2}$$



$st \rightarrow 0$
 $mid \rightarrow 0$
 $en \rightarrow 1$
 $K=0$
return

$(4 \ 5 \ 6 \ 7) \ (0 \ 1 \ 2)$
 $st \quad \quad \quad mid \quad \quad \quad en$

$\frac{st}{4} \times \frac{mid}{7} \leq K \leq 7$
 $= 0$

$K=0$
 $(0 \ 1 \ 2)$
 $st \quad \quad \quad mid \quad \quad \quad en$

$st \quad \quad \quad mid$
 $(0 \leq K \leq 1)$
 $? \leq 0 \leq 1$